Serial Communications with SPI on the MSP432

ECE 2564, Embedded Systems

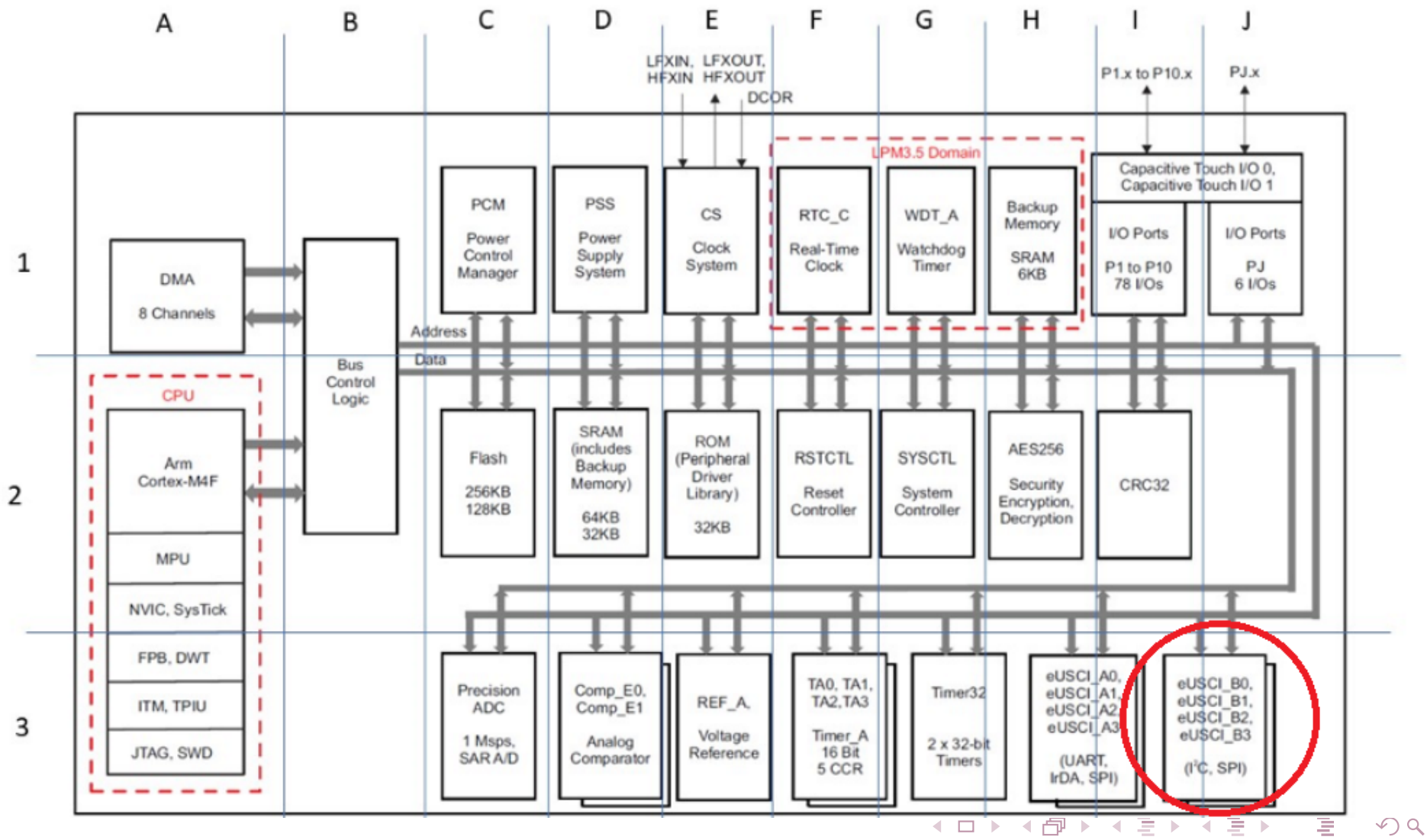
Fall 2025

Overview

SPI on the MSP432

- Look at the hardware
  - Board level, chip level, SPI level
- Look at the registers
- Look at what needs to be programmed

SPI on the MSP Functional Block Diagram

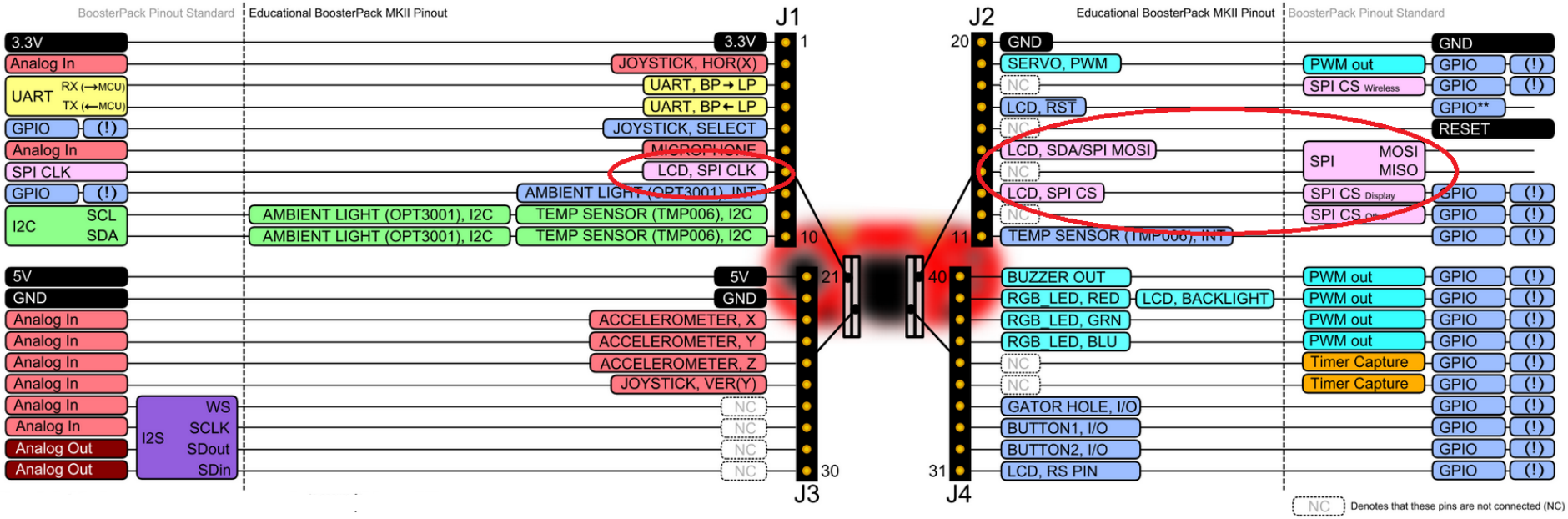


SPI on the MSP Functional Block Diagram

- Several eUSCI modules are capable of SPI communication
- We will only use eUSCI_B
- Each can act as a master or a slave
- Data rates up to 16 Mbps are supported

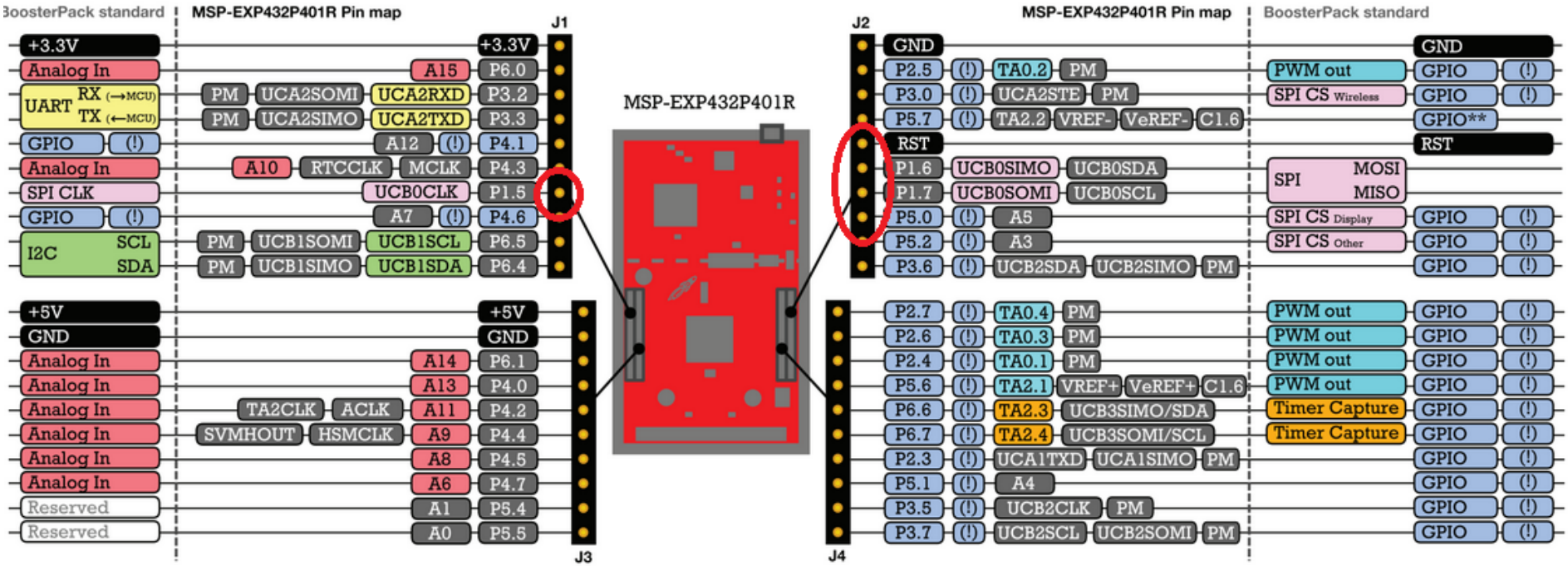
Where is SPI Used on the BoosterPack? Think PINK

The LCD is supported by SPI

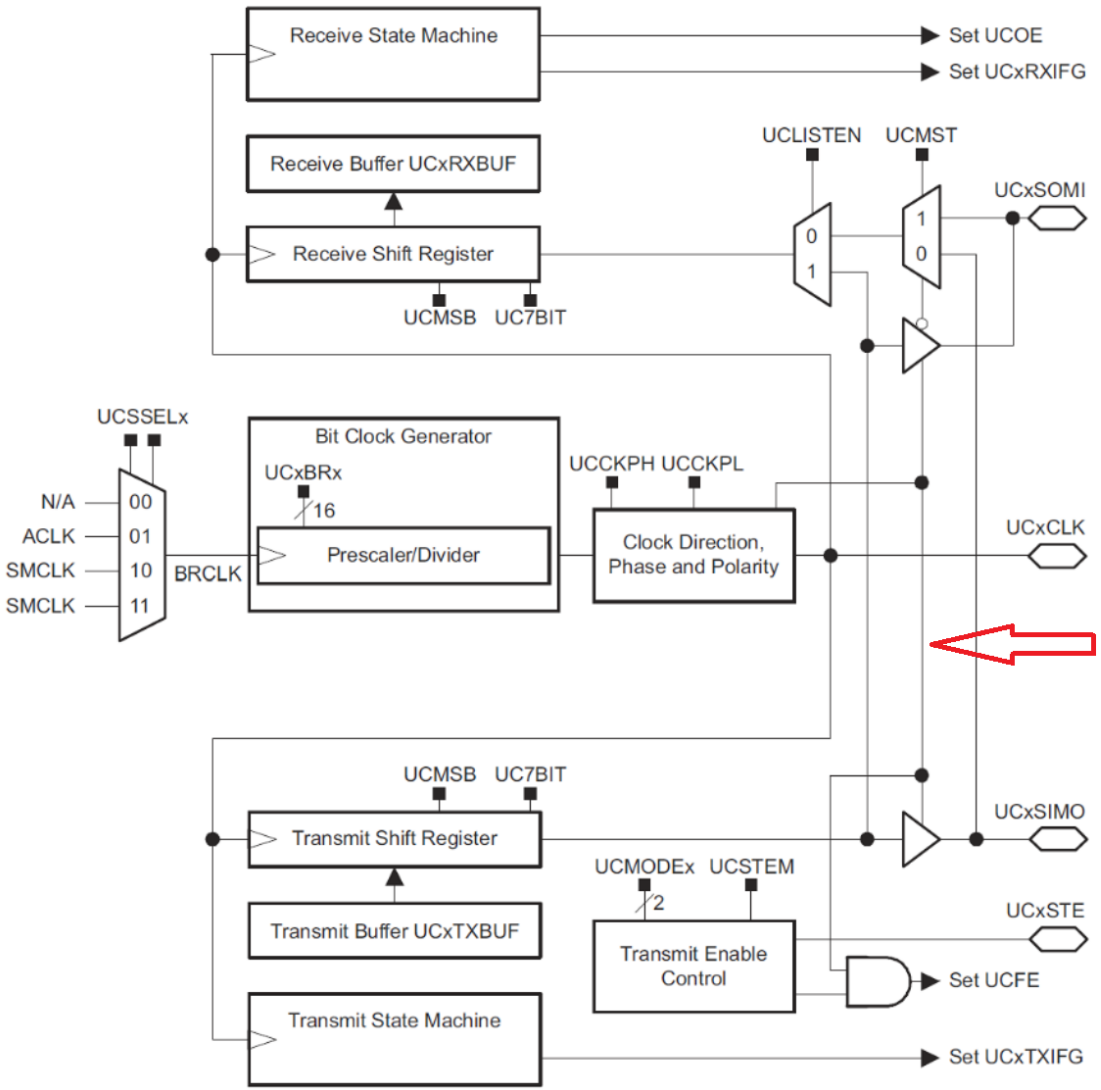


Ports/Pins Related to SPI on the MSP432 for the LCD

SPI CLK: P1.5, SPI MOSI: P1.6, SPI CS: P5.0, SPI MISO: Not used



The SPI (eUCSCI) Block Diagram



- Inside the eUCSCI
- What's new?
  - UCCKPH
  - UCCKPL
  - Tristates
  - Muxes

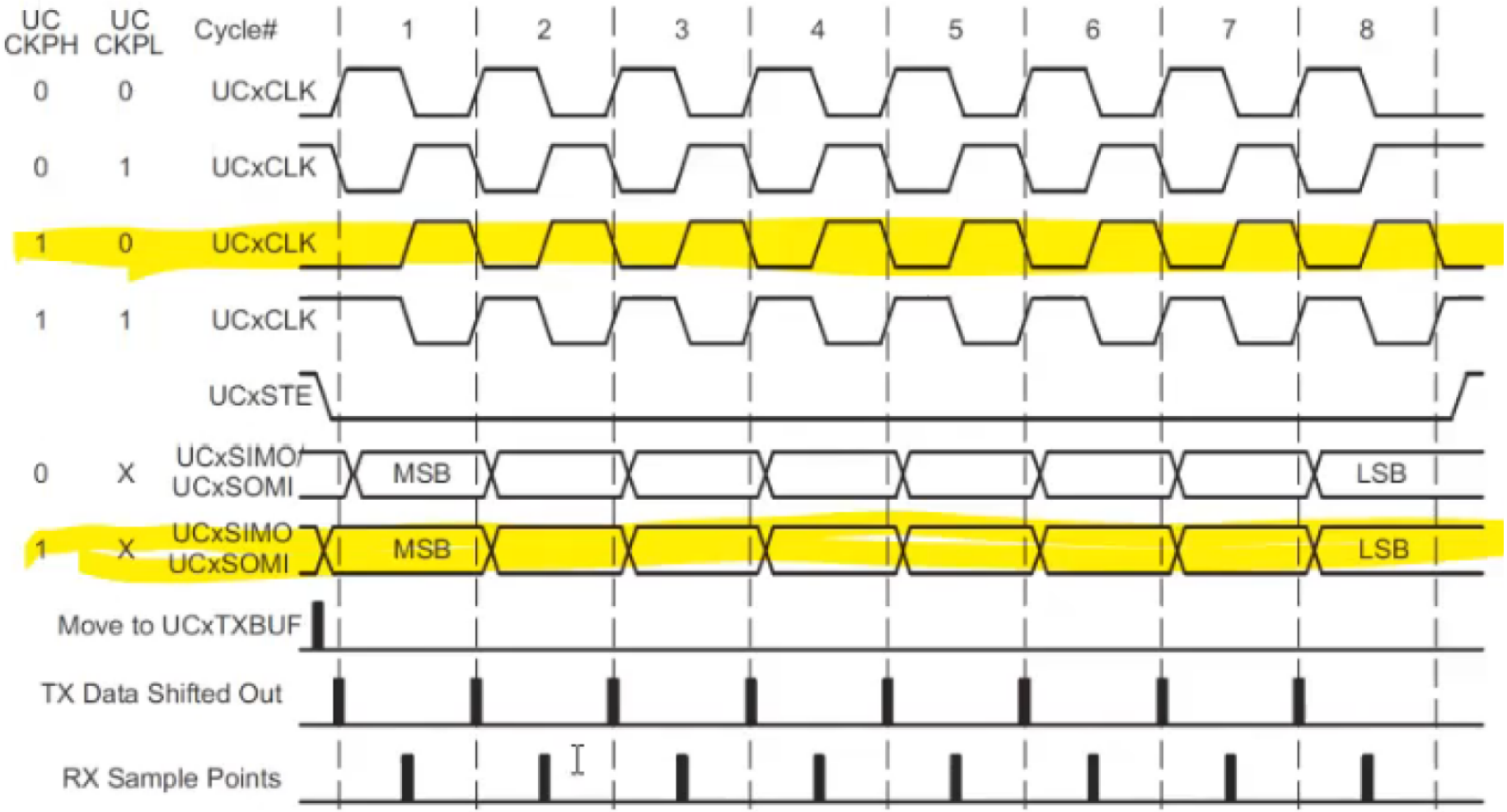
Enable to the tristate

- 1: SOMI to RxShiftReg
- TxShiftReg to SIMO

SPI Send/Receive

- Lots of components look familiar to previous protocols
  - Send/Receive shift register
  - Send/Receive buffer
  - Clock multiplexer
  - Clock prescalar
- Some components are new
  - Clock direction, phase and polarity control
  - Extra tri-state buffers and multiplexers close to pins — control SOMI (MISO)/SIMO (MOSI)

SPI Timing Revisited



We've Been Using SPI With the Graphics Screen

https://github.com/ECE2564-VT/basic_example_graphics.git

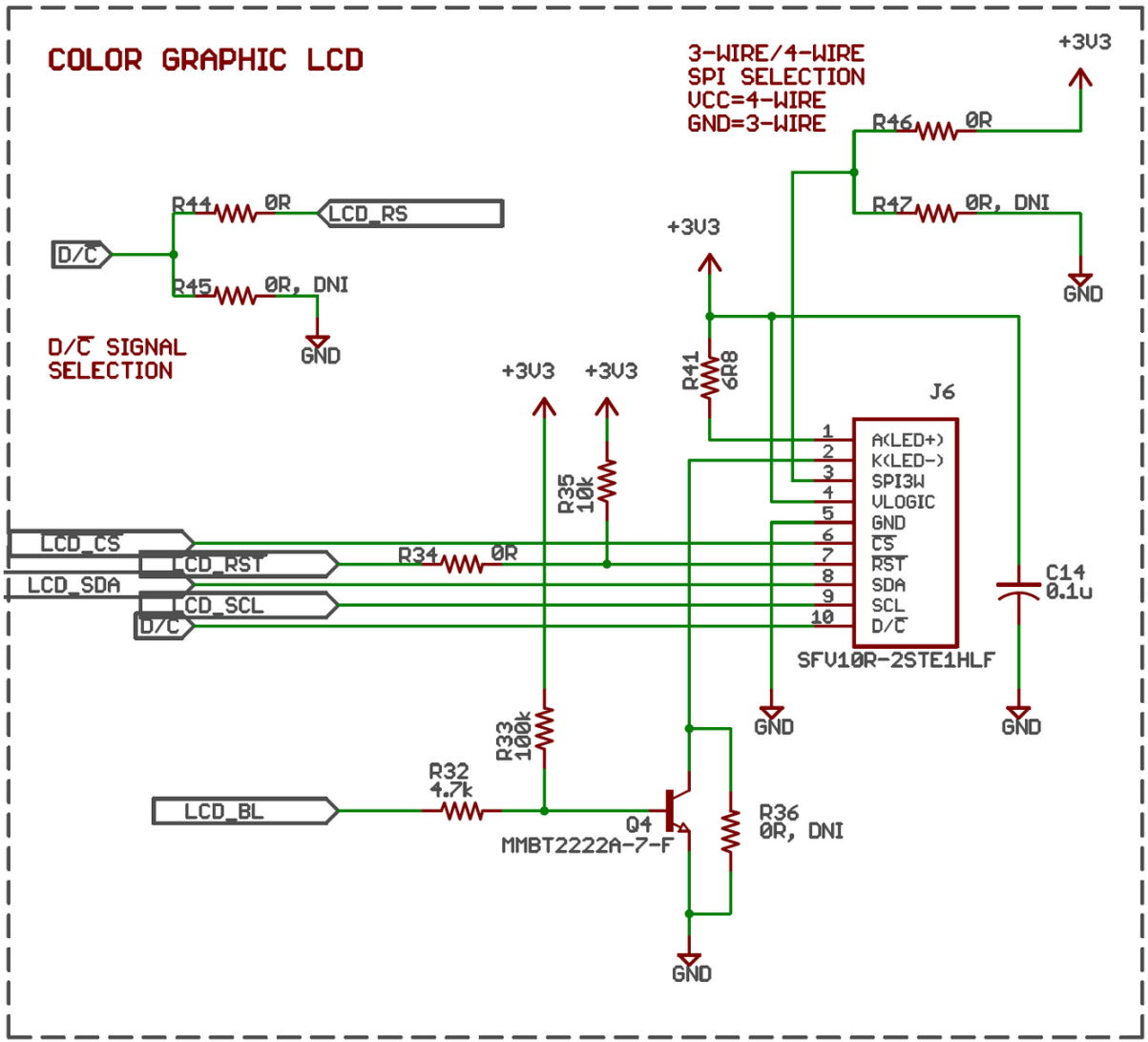
Table 7. Color LCD Pinout

BoosterPack Plug-in Module Header Connection	Pin Function
J1.7	LCD SPI clock
J2.13	LCD SPI chip select
J2.15	LCD SPI MOSI
J4.31	LCD reset pin
J4.39(1)	LCD backlight

(1) Pin is multiplexed with the RGB LED red channel pin through the jumper header J5.

- CS: Chip Select (active low)
- RST: Reset (active low)
- DC: Date (1), Command (0)

The LCD Color Graphic SPI Circuit



Let's Look at Code for Graphics

Which Protocol to Use?

	I2C	SPI	UART
On one PCB or stacked PCBs	☑	☑	☐
between boards	☐	☐	☑

I2C & SPI require a clock
Synchronize clock –
short distance

Select the communication protocols that match the relative cost

	I2C	SPI	UART
cheap	☑	☐	☑
more expensive	☐	☑	☐

UART – cheapest
I2C – 2 pins
SPI – send/receive at same time

Select the communication protocols that match the relative speed

	I2C	SPI	UART
slow (~kbit/sec)	☑	☐	☑
faster (~Mbit/sec)	☐	☑	☐

SPI – no ACK, no slave address
UART – start/stop, async
I2C – need address, etc.

Comparing Communication Protocols

	I2C	SPI (4-wire SPI)	UART
Synchronous vs. Asynchronous	*Synchronous*	*Synchronous*	*Asynchronous*
Serial vs. Parallel	*Serial*	*Serial*	*Serial*
Half- vs. full-duplex	*Half*	*Full*	*Full*
Choosing a slave	*Send the address of the slave*	*SS signal chooses the slave*	*NA: Master-less scheme*
Starting communication	*Start condition (a pattern within SCL and SDA)*	*SS again decides the communication start*	*Start bit (idle bits are 1 and a falling edge starts the comm.)*
Stopping communication	*Stop condition*	*SS goes back to 1*	*Stop bit*
Read vs. Write selection	*A bit in the package tells the slave if this is a read or write*	*Read and write happen at the same time.*	*Read and write are independent*
Acknowledgement/ data checking	*An explicit ACK bit driven by the receiver*	*There is none*	*Optional parity bit sent by the sender*

Comparing Communication Protocols

When can the transmitter update the data?	When the clock is low	It is configurable (on the rising or falling edge of the clock)	Once a full bit/baud has complete.
When can the data be sampled?	When the clock is high	It is configurable (opposite of when the data can change)	In the middle of bit/baud.
# of pins	2	4 pins if you have one slave; 3+number of slaves (if master has direct connection to each slave)	2 pins for TX and RX -- plus an extra for ground
# of slaves	Multiple (128 logical limit, 400 pf capacitance limit)	Multiple (limited to the number of SS pins - also limited for capacitance/speed)	There are no slaves!
# of masters	Supports multiple masters	Support only one master	There are no masters.

SPI Summary

- The Serial Peripheral Interface is a popular way to connect several devices at the *board* level
  - Synchronous, full-duplex connection
  - SCLK, MOSI, MISO, $\overline{\text{SS}}$
  - Normally uses a separate select line for each peripheral
  - Direct configuration versus Daisy Chain configuration
  - CPOL and CPHA configurable